

Homework 1

6501 Online Optimization and Learning in Games (Fall 2026)

Deadline: 11:59pm, September 20, 2026

A L^AT_EX template is available at <https://www.overleaf.com/read/dxwndthbvgjb#f24e32>.

1 Adaptive learning rates for exponential weights

Consider the experts problem with $N \geq 2$ experts and loss vectors $\ell_t \in [0, 1]^N$. At round t , the learner chooses $p_t \in \Delta_N$, observes ℓ_t , and suffers loss $\langle p_t, \ell_t \rangle$. Let $L_t(i) = \sum_{s=1}^t \ell_s(i)$ denote the cumulative loss of expert i through round t , with $L_0(i) = 0$. We would like to adapt the learning rate to the observed squared losses without knowing their cumulative weighted sum or T in advance.

Initialize $p_1(i) = 1/N$. After observing ℓ_t , set

$$\eta_t = \sqrt{\frac{\ln N}{\ln N + \sum_{s=1}^t \sum_{i=1}^N p_s(i) \ell_s(i)^2}}$$

and use exponential-weights update with time-varying temperature:

$$p_{t+1}(i) = \frac{\exp(-\eta_t L_t(i))}{\sum_{j=1}^N \exp(-\eta_t L_t(j))}.$$

Define

$$\Phi_t(\lambda) = \frac{1}{\lambda} \ln \left(\frac{1}{N} \sum_{i=1}^N \exp(-\lambda L_t(i)) \right), \quad \text{for } \lambda > 0.$$

(a) Prove the one-step inequality

$$\langle p_t, \ell_t \rangle \leq \Phi_{t-1}(\eta_{t-1}) - \Phi_t(\eta_{t-1}) + \eta_{t-1} \sum_{i=1}^N p_t(i) \ell_t(i)^2.$$

and show that

$$\text{Regret} \leq \frac{\ln N}{\eta_T} + \sum_{t=1}^T \eta_{t-1} \sum_{i=1}^N p_t(i) \ell_t(i)^2 + \sum_{t=1}^T (\Phi_t(\eta_t) - \Phi_t(\eta_{t-1})).$$

(Hint: Follow the proof on Page 8 [here](#).)

(b) Prove that $\Phi_t(\lambda)$ is nondecreasing in $\lambda > 0$. Deduce that the final sum in part (a) is nonpositive. (Hint: You may first prove $\Phi_t(\lambda) = -\min_{p \in \Delta_N} \{ \langle p, L_t \rangle + \frac{1}{\lambda} \text{KL}(p, \mathbf{1}/N) \}$, where $\mathbf{1}/N$ denotes the uniform distribution over Δ_N .)

(c) Combine parts (a)–(b) with the specified learning-rate schedule to prove that, for every $T \geq 1$,

$$\text{Regret} \leq C_0 \left(\ln N + \sqrt{(\ln N) \sum_{t=1}^T \sum_{i=1}^N p_t(i) \ell_t(i)^2} \right),$$

where C_0 is a universal constant.

2 Comparator-dependent regret bounds

Consider the experts problem with N experts and loss vectors $\ell_t \in [0, 1]^N$. At each round t , the learner chooses $p_t \in \Delta_N$, observes ℓ_t , and suffers loss $\langle p_t, \ell_t \rangle$. As in Problem 1, let $L_t(i) = \sum_{s=1}^t \ell_s(i)$ denote the cumulative loss of expert i through round t , with $L_0(i) = 0$. For each expert $i \in [N] = \{1, \dots, N\}$, define the instantaneous regret compared to expert i as

$$r_t(i) = \langle p_t, \ell_t \rangle - \ell_t(i).$$

The cumulative regret compared to expert i is

$$\text{Regret}(i) = \sum_{t=1}^T r_t(i) = \sum_{t=1}^T \langle p_t, \ell_t \rangle - L_T(i).$$

Thus, $\text{Regret}(i)$ is the learner's total loss minus the total loss of expert i , while $\text{Regret} = \max_{i \in [N]} \text{Regret}(i)$ is the regret compared to the best fixed expert in hindsight.

(a) **(Refined analysis for exponential weights)** Recall that exponential weights uses

$$p_{t+1}(i) = \frac{\exp(-\eta L_t(i))}{\sum_{j=1}^N \exp(-\eta L_t(j))}.$$

For any fixed $0 < \eta \leq 1$, prove that this algorithm ensures

$$\text{Regret} \leq \frac{\ln N}{\eta} + \eta \sum_{t=1}^T \sum_{i=1}^N p_t(i) r_t(i)^2.$$

(Hint: Does subtracting the same scalar from every expert's loss change the exponential-weights distributions? Apply the analysis from class to suitably shifted losses.)

(b) **(A comparator-dependent bound)** The second-order term in the regret in part (a) averages over the learner's distribution. We now seek a bound depending only on the deviations of the comparator expert. Define

$$C_t(i) = \sum_{s=1}^t r_s(i)^2, \quad C_0(i) = 0.$$

For any fixed $0 < \eta \leq \frac{1}{2}$, consider the update

$$p_{t+1}(i) = \frac{\exp(-\eta L_t(i) - \eta^2 C_t(i))}{\sum_{j=1}^N \exp(-\eta L_t(j) - \eta^2 C_t(j))},$$

Prove that this algorithm ensures, simultaneously for all $i \in [N]$,

$$\text{Regret}(i) \leq \frac{\ln N}{\eta} + \eta \sum_{t=1}^T r_t(i)^2.$$

(Hint: You may use $\exp(-x - x^2) \leq 1 - x$ for $|x| \leq \frac{1}{2}$.)

Remark. This result is a preparation for topics to be discussed in the class. Combining Problem 2(a) with the learning-rate scheduling idea in Problem 1 gives a bound of order

$$\text{Regret} \lesssim \ln N + \sqrt{(\ln N) \sum_{t=1}^T \sum_{i=1}^N p_t(i) r_t(i)^2}.$$

For Problem 2(b), however, it is unclear how a single learning-rate schedule can give the comparator-dependent bound of

$$\text{Regret}(i) \lesssim \ln N + \sqrt{(\ln N) \sum_{t=1}^T r_t(i)^2} \tag{1}$$

for all experts i simultaneously, because the best learning rate suggested by the bound can differ across comparators. It turns out that with a more involved algorithm design, (1) is indeed achievable for all i simultaneously, up to additional logarithmic factors. Such guarantees will be crucial for the adaptive regret bounds we discuss later in the course.

3 Online Mirror Descent

Let $\mathcal{X} \subseteq \mathbb{R}^d$ be a convex set. At round t , the learner chooses $x_t \in \mathcal{X}$, suffers loss $f_t(x_t)$, and observes $g_t = \nabla f_t(x_t)$. For a differentiable convex function h , define its Bregman divergence by

$$D_h(u, x) = h(u) - h(x) - \langle \nabla h(x), u - x \rangle.$$

Given a convex regularizer ψ and a fixed learning rate $\eta > 0$, Online Mirror Descent (OMD) uses

$$x_{t+1} \in \operatorname{argmin}_{x \in \mathcal{X}} \left\{ \langle x, g_t \rangle + \frac{1}{\eta} D_\psi(x, x_t) \right\}. \quad (2)$$

(a) Let $F : \mathcal{X} \rightarrow \mathbb{R}$ be convex and differentiable, and let $x^* \in \operatorname{argmin}_{x \in \mathcal{X}} F(x)$. Prove that for every $u \in \mathcal{X}$,

$$F(x^*) \leq F(u) - D_F(u, x^*).$$

(b) Apply part (a) to the OMD objective (2) to show that, for every $u \in \mathcal{X}$,

$$\eta \langle x_{t+1} - u, g_t \rangle \leq D_\psi(u, x_t) - D_\psi(u, x_{t+1}) - D_\psi(x_{t+1}, x_t). \quad (3)$$

Derive the following bound on the regret compared to u :

$$\begin{aligned} \sum_{t=1}^T (f_t(x_t) - f_t(u)) &\leq \sum_{t=1}^T \langle x_t - u, g_t \rangle \\ &\leq \frac{D_\psi(u, x_1)}{\eta} + \sum_{t=1}^T \left(\langle x_t - x_{t+1}, g_t \rangle - \frac{D_\psi(x_{t+1}, x_t)}{\eta} \right). \end{aligned} \quad (4)$$

(c) Suppose ψ is μ -strongly convex with respect to a norm $\|\cdot\|$, and $\|\cdot\|_*$ is its dual norm. Use (4) to establish, for every $u \in \mathcal{X}$,

$$\sum_{t=1}^T (f_t(x_t) - f_t(u)) \leq \frac{D_\psi(u, x_1)}{\eta} + \frac{\eta}{2\mu} \sum_{t=1}^T \|g_t\|_*^2. \quad (5)$$

(d) Specialize to the expert setting where $\mathcal{X} = \Delta_N$ and $f_t(x) = \langle x, \ell_t \rangle$. Take the negative-entropy regularizer

$$\psi(x) = \sum_{i=1}^N x(i) \ln x(i).$$

Verify that OMD with this regularizer gives the exponential-weights update $x_{t+1}(i) \propto x_t(i) \exp(-\eta \ell_t(i))$.

(e) Let $\psi(x) = \frac{1}{2} \|x\|_2^2$. Verify that OMD with this regularizer recovers projected gradient descent:

$$x_{t+1} = \Pi_{\mathcal{X}}(x_t - \eta g_t), \quad \Pi_{\mathcal{X}}(y) = \operatorname{argmin}_{x \in \mathcal{X}} \|x - y\|_2^2.$$

Here $\Pi_{\mathcal{X}}$ denotes Euclidean projection onto $\mathcal{X}$. Suppose $\|g_t\|_2 \leq G$ for all t , and let $D = \sup_{x, u \in \mathcal{X}} \|x - u\|_2 < \infty$ be the diameter of $\mathcal{X}$. Use (5) and choose η to obtain $\text{Regret} \leq DG\sqrt{T}$.

4 FTPL with Gaussian perturbations

Consider the experts problem with $N \geq 2$ and a loss sequence $\ell_1, \dots, \ell_T \in [0, 1]^N$ fixed in advance. As before, let $L_t(i) = \sum_{s=1}^t \ell_s(i)$ and $L_0(i) = 0$. For a fixed $\eta > 0$, Follow the Perturbed Leader (FTPL) draws an independent $Z_t \sim \mathcal{N}(0, I_N)$ on each round and chooses

$$i_t = \operatorname{argmin}_{i \in [N]} \left\{ L_{t-1}(i) + \frac{Z_t(i)}{\eta} \right\}.$$

Let $p_t(i) = \Pr(i_t = i)$, and define p_{T+1} by the same rule using L_T . Our goal is to prove

$$\mathbb{E}[\text{Regret}] = \mathbb{E} \left[\sum_{t=1}^T \ell_t(i_t) \right] - \min_{i \in [N]} L_T(i) = \mathcal{O}(\sqrt{T \ln N}).$$

Rather than using a total-variation bound on the full perturbation distribution, we will refine the stability analysis by considering an auxiliary update that adds only one expert's loss. You may use the following decomposition from the FTPL analysis, where $Z \sim \mathcal{N}(0, I_N)$:

$$\mathbb{E}[\text{Regret}] \leq \frac{\mathbb{E}[\max_{i \in [N]} Z(i) - \min_{i \in [N]} Z(i)]}{\eta} + \sum_{t=1}^T \langle p_t - p_{t+1}, \ell_t \rangle. \quad (6)$$

(a) Fix an expert i and define the auxiliary distribution $p_{t+1}^{(i)} \in \Delta_N$ by

$$p_{t+1}^{(i)}(j) = \Pr \left(j = \operatorname{argmin}_{k \in [N]} \left\{ L_{t-1}(k) + \ell_t(i) \mathbb{I}\{k = i\} + \frac{Z_t(k)}{\eta} \right\} \right).$$

Show that $p_{t+1}^{(i)}(i) \leq p_{t+1}(i)$. Then, by comparing Gaussian densities, prove

$$p_{t+1}^{(i)}(i) = \mathbb{E} \left[\mathbb{I}\{i_t = i\} \exp \left(\eta \ell_t(i) Z_t(i) - \frac{\eta^2 \ell_t(i)^2}{2} \right) \right].$$

The expectation uses the same Z_t that defines i_t . (*Hint: The auxiliary update increases only expert i 's cumulative loss. For the density comparison, use $\varphi(z - a) = \varphi(z) \exp(az - a^2/2)$, where φ is the standard normal density.*)

(b) Use part (a) to establish the refined stability bound

$$\langle p_t - p_{t+1}, \ell_t \rangle \leq \eta \mathbb{E}[\|Z\|_\infty] + \frac{\eta^2}{2}. \quad (7)$$

Apply (6) and choose η to conclude $\mathbb{E}[\text{Regret}] = \mathcal{O}(\sqrt{T \ln N})$. You may use the Gaussian maximum bound $\mathbb{E}[\|Z\|_\infty] \leq \sqrt{2 \ln(2N)}$. (*Hint: Use $e^x \geq 1 + x$ and sum the resulting expert-wise inequalities after multiplying by $\ell_t(i)$. Exactly one indicator $\mathbb{I}\{i_t = i\}$ is nonzero.*)